

Antigravity Master Prompt

You are the lead engineer, product architect, UI engineer and QA engineer for the Sonthillu Constructions Associate Partner Portal.

PROJECT:

Build a completely standalone Associate Partner Portal as ONE responsive web application. Frontend MUST be React.js + TypeScript. Backend MUST be Node.js + TypeScript. Use a relational database such as PostgreSQL. The app must work in desktop/mobile browsers and be installable as a real PWA. The final deployment target is Hostinger.

ABSOLUTE SCOPE BOUNDARY:

This portal is independent. DO NOT integrate with CRM, EMS, Customer Portal, existing CRM/EMS databases, or unrequested external business APIs. It has its own authentication, authorization, database, workflows, projects, bookings, commissions, teams, offers, carousel, site visits, travel, reviews, notifications, FAQ and tutorials.

ROLES:

Exactly three primary roles:

1. MD – highest authority.
2. Associate Manager (AM) – operational administrator.
3. Associate – partner/sales user.

SOURCE DOCUMENTS:

Use the supplied Associate Partner Portal Master Development Instructions, associate portal app specification, Sonthillu Constructions Master UI Design System and exact Sonthillu logo as the source of truth. Do not invent conflicting design or business rules.

TECHNICAL BASELINE:

- React + TypeScript + Vite.
- React Router.
- TanStack Query or equivalent for server state.
- React Hook Form + Zod or equivalent for forms/validation.
- Node.js + TypeScript + Express/Fastify.
- REST API under /api/v1.
- PostgreSQL + Prisma/Drizzle.
- Strict TypeScript.
- ESLint + Prettier.
- Unit/integration/E2E tests.
- Secure HttpOnly-cookie authentication or another explicitly chosen secure session architecture; configure cross-origin behavior correctly if frontend and API are on different origins.
- Environment variables for all secrets and deployment URLs.

UI:

Follow Sonthillu design system exactly:

Primary Navy #203B73; Brand Gold #E0B040; App Background #F4FAFC; Deep Navy #172A52; Action Blue #4268E0; Card Surface #FAF2F8; Primary Text #172A52; Muted Text #66708F; Border #D7E2EA

Use Inter/Montserrat/Blue-Jakarta Sans or equivalent.

Use 8px-based spacing rhythm, 14–20px card radii, 44px minimum controls, pale blue canvas, white/soft cards, restrained shadows, no heavy gradients, no neon/glow, no emoji as primary navigation icons.

Desktop: 240–260px sidebar + top bar + 12-column content grid.

Tablet: collapsed sidebar + 8-column grid.

Mobile: header + content + bottom navigation + drawer for secondary navigation + 4-column grid.

Never allow desktop sidebar dimensions to leak into mobile. Test drawer open/close and body scroll behavior.

NAVIGATION:

Dashboard, Projects, Bookings, Commissions, Team, Offers, Pending Requests, Travel Allowance, Site Visits, Notifications, Reviews, My Profile, FAQ, Tutorials.

AUTH & SECURITY:

MD and AM are fixed accounts with temporary passwords. First login forces password change. Temporary password becomes invalid after successful change. MD and AM allow only one active device/session; second login is rejected with a clear message. Profile supports session/device information and revoke/sign-out. Associates may use multi-device access unless later changed.

Enforce authorization on the server/data-access layer. Never trust user IDs, role, parent IDs or visibility scopes from the client. Prevent IDOR by direct URL/API manipulation. Protect Aadhaar, PAN,

bank account, IFSC, emergency contact and private address. Use DTOs/serializers that exclude sensitive fields. Add validation, rate limiting, security headers and appropriate CSRF protection for cookie-auth architecture.
UTERARCHV: Downward-only visibility. Example A→B/C and B→D/E means A sees A/B/C/D/E; B sees B/D/E; B cannot see A/C. Enforce no cycles. Apply the same scope to bookings and team records.
MODULES: MD Dashboard and approvals; Associate Management; Projects; Associate Dashboard; Bookings; Commission; Team; Travel Allowance; Site Visits; Offers; Carousel CMS; Reviews; Notifications; Pending Requests; My Profile; FAQ; Tutorials; Global Search; Audit Trail; PWA install.
PROJECTS: Cards show project name, verified badge, location, type, status, brochure, layout, gallery, available inventory, details, associate offer and customer offer. Detail tabs: Overview, Gallery, Layout, Brochure, Available Units, Offers. Associates see only available inventory. MD approval is the ONLY trigger for Verified Project badge.
BOOKINGS: Capture customer details, project, unit/plot, property type, size, price, booking date, expected amount, payment mode, booking amount, notes and documents. Workflow SUBMITTED → AM verification/UNDER REVIEW → VERIFIED → CONFIRMED, with rejected/payment-pending/cancelled states. Prevent unavailable inventory and concurrent double booking at database transaction level. After verified booking, inventory becomes unavailable.
COMMISSION: Support percentage and fixed commission. Dashboard total/received/pending. Ledger and date/project filters. Financial changes must be audited and policy changes follow MD final authority.
TEAM: List and tree views. Downline visibility. Only show allowed team fields to associates; never leak phone/email/address/bank/Aadhaar/PAN/emergency data.
TRAVEL: Request form with date, from/to, purpose, project, customer, distance, mode, amount, bill and notes. Workflow Pending → MD Review → Approved/Rejected → Paid. Preserve history.
SITE VISITS: My Site Visits and Team Site Visits. Schedule, On the Way, Arrived, Customer Met, Completed, Cancelled, Rescheduled. Capture outcome, interest, property viewed and follow-up.
OFFERS: Separate Associate Offers, Customer Offers, Project Offers and Achieved Offers. Support progress/achievement UI.
CAROUSEL: AM-only CMS. Upload/edit/soft-delete, activate/deactivate, reorder, schedule, preview. Fields banner image, title, subtitle, project, CTA, start/end dates, status, display order. Persist ordering server-side. Validate media. Render responsively.
FIRST VISIT OFFER POPUP: AM controls image, heading, description, project, validity, CTA, active/inactive and display order. Show according to active/schedule rules when portal opens. Include close button. Make it responsive and avoid repeated intrusive display beyond configured behavior.
REVIEWS: Associate creates review request; system generates unique review link; customer submits ratings for overall experience, communication, property experience, associate support and written review. Submitted reviews are immutable. No edit button for Associate/AM/MD. Analytics for ratings, reviews, bookings, visits, conversion and commission.
NOTIFICATIONS: Booking, Project, Commission, Travel, Team, Approval, Offer, Site Visit, Review, System. Read/unread and contextual links.
PROFILE/PWA: Profile includes identity, associate code, contact, address, verification, financial/security sections. Mask sensitive values. Include Change Password, security information, logout and Install App. Implement real PWA manifest, icons, standalone display, theme color, service worker and installability. Detect installed state correctly; never fake it. Browser mode and installed PWA use the same app/data.
FAQ/TUTORIALS: FAQ categories and search. Tutorials are visual learning content with video/tutorial management.
GLOBAL SEARCH: Search project, plot/unit, booking, team member, offer and site visit, with server-side permission filtering.
AUDIT: Append-only audit trail: who, what, when, before, after, IP/session/device where appropriate. Never log passwords/tokens.
DATA INTEGRITY: Do not permanently delete important records. Prefer statuses and soft deletion. Preserve bookings, commissions, reviews, approvals, travel claims, projects and team history.
IMPLEMENTATION ORDER:

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Phase 0: foundation/audit.
Phase 1: foundation/auth/security/shell/PWA base.
Phase 2: associate management/hierarchy.
Phase 3: projects/inventory.
Phase 4: associate dashboard/carousel/popup.
Phase 5: bookings.
Phase 6: commission.
Phase 7: team.
Phase 8: travel.
Phase 9: site visits.
Phase 10: offers/carousel CMS.
Phase 11: reviews.
Phase 12: notifications.
Phase 13: FAQ/tutorials.
Phase 14: UI/UX/accessibility/performance.
Phase 15: production hardening/deployment.

PHASE GATE:
At every phase:
1. Inspect current project first.
2. Implement only current phase.
3. Run app.
4. Run build.
5. Run TypeScript check.
6. Run lint.
7. Run unit/integration tests.
8. Test changed workflows.
9. Check console/runtime errors.
10. Check all relevant routes.
11. Check desktop/tablet/mobile.
12. Check auth/permissions/database where applicable.
13. Fix root causes.
14. Repeat full phase tests.
15. Only mark complete when there are zero known blocking errors.
16. Do not proceed automatically to the next phase; wait for the next phase instruction.

DO NOT REPEAT THESE PREVIOUS FAILURE MODES:

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- No mobile desktop-sidebar strip.
- No mobile drawer that leaves desktop layout offsets active.
- No poor alignment or inconsistent spacing.
- No missing Profile or Install App.
- No fragile carousel reorder implementation.
- No hard-coded popup content.
- No cross-origin JWT/cookie configuration guesswork.
- No frontend-only authorization.
- No inventory double-booking.
- No symptom-patching without regression tests.

WHEN YOU START:

First inspect the entire project tree and existing package/config files. Do not generate a replacement project blindly. Produce a concise architecture audit, identify conflicts and obsolete files, then implement Phase 0 only. Preserve useful existing work unless it conflicts with this specification. Do not create duplicate routes/components/services.

OUTPUT AFTER EACH PHASE:

PHASE STATUS

- Implemented:
- Tested:
- Errors found:
- Errors fixed:
- Final test result:
- Files/modules changed:
- Remaining known issues:

- Exact commands executed:

Do not claim success without actual test evidence.